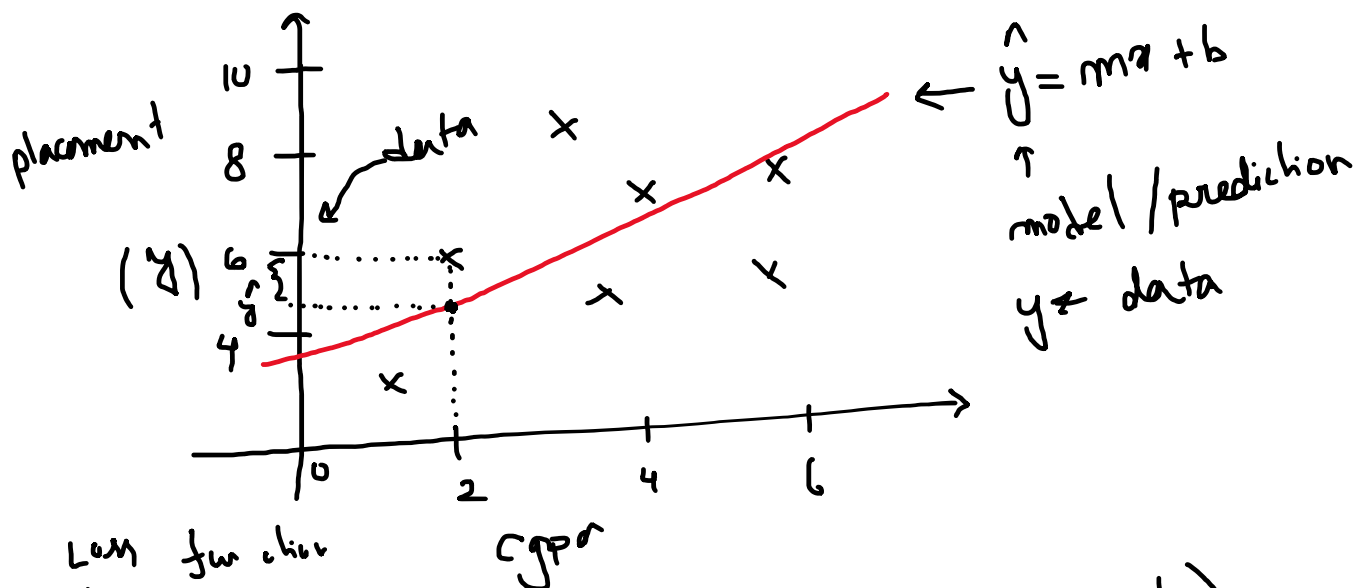
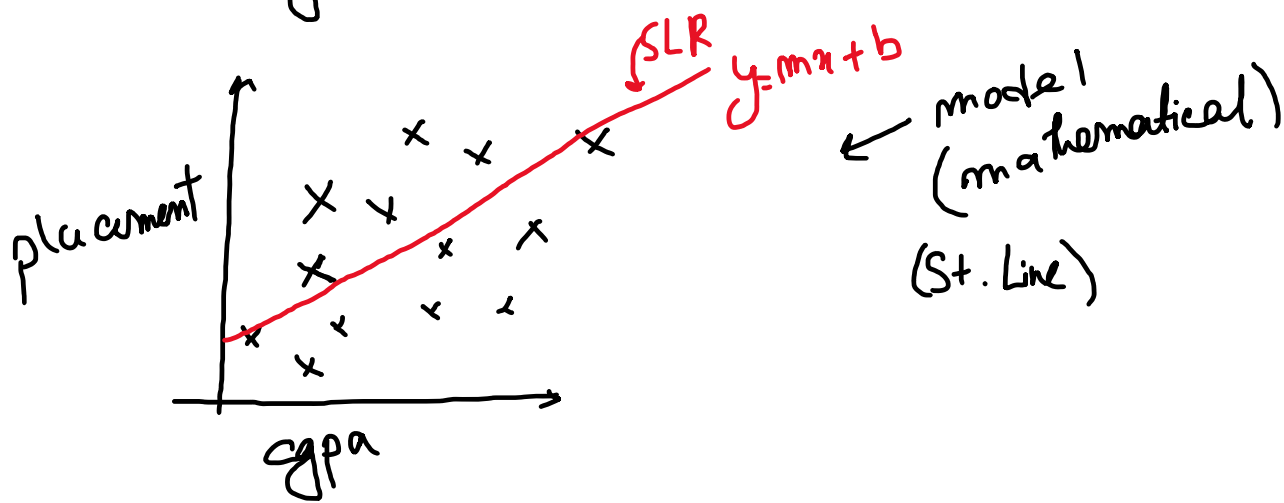


LR → Line

→ Simple
→ Multiple
→ Polynomial

cgpa	place
7.7	1.1
5.0	0
...	...
...	...



Loss function

$$P_i = y_i - \hat{y}_i = y_i - (mx_i + b) = (y_i - mx_i - b)$$

$$\rightarrow L = y_i - \hat{y}_i = (y_i - mx_i - b)$$

placement(y)

→ L u u

row = 4 row = 4 placement(y)

1	8	8L ←
2	9	8.9
3	4	0
4	7	5.2

#row = 4

row 1 → $y_1 - \hat{y}_1$
row 2 → $y_2 - \hat{y}_2$

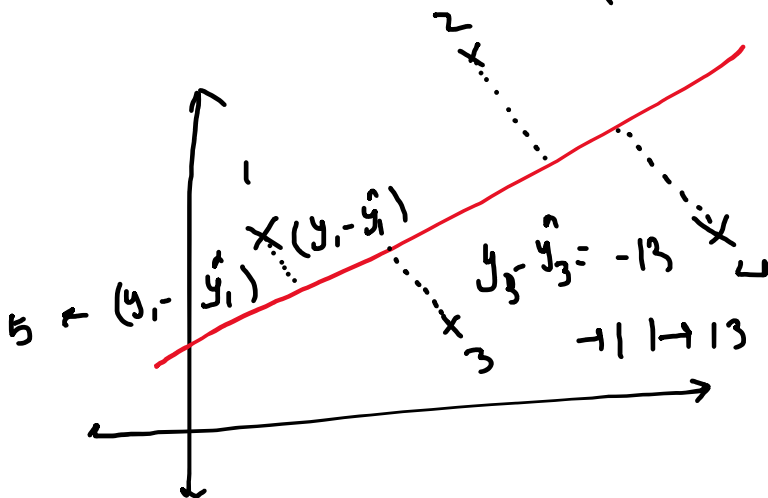
row 3 → $y_3 - \hat{y}_3$
row 4 → $y_4 - \hat{y}_4$

$$\begin{aligned} \#row = n \\ + &= (y_1 - \hat{y}_1) + (y_2 - \hat{y}_2) + (y_3 - \hat{y}_3) + (y_4 - \hat{y}_4) \\ &+ \dots + (y_n - \hat{y}_n) \\ &\Rightarrow \sum_{i=1}^n (y_i - \hat{y}_i) \end{aligned}$$

$$C = \frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)$$

(HVE)

← MRE ←
metric



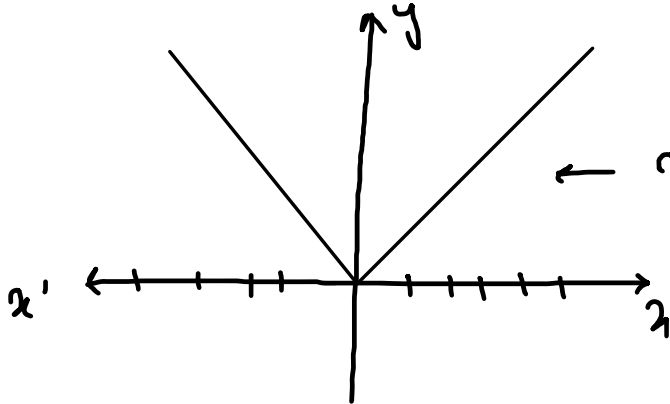
$$\begin{aligned} 5 + 13 &= 18 \\ 5 - 13 &= -8 \end{aligned}$$

function

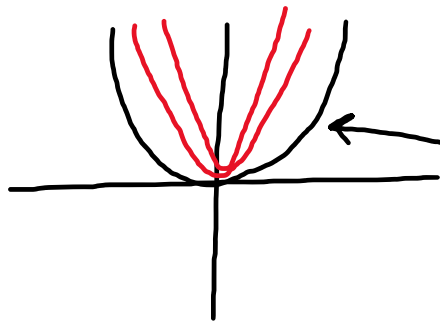
↓

$$C = \frac{1}{n} \sum_{i=1}^m |(y_i - \hat{y}_i)|$$

mod function



cont
at $x=0$ $\frac{dy}{dx} \times$

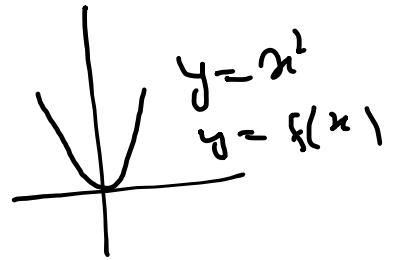


$$f(x) = x^n$$

$$n \geq 2m$$

$m \rightarrow (0, \infty)$

$$y = x^2$$



$$L = (y_i - \hat{y}_i)^2$$

$$C = \frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2 = \frac{1}{n} \sum_{i=1}^n (y_i - mx_i - b)^2$$

$$C = f(m, b)$$

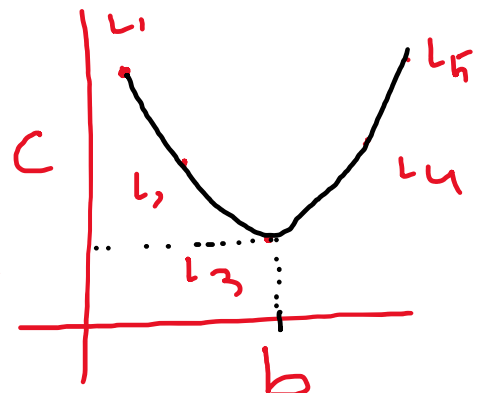
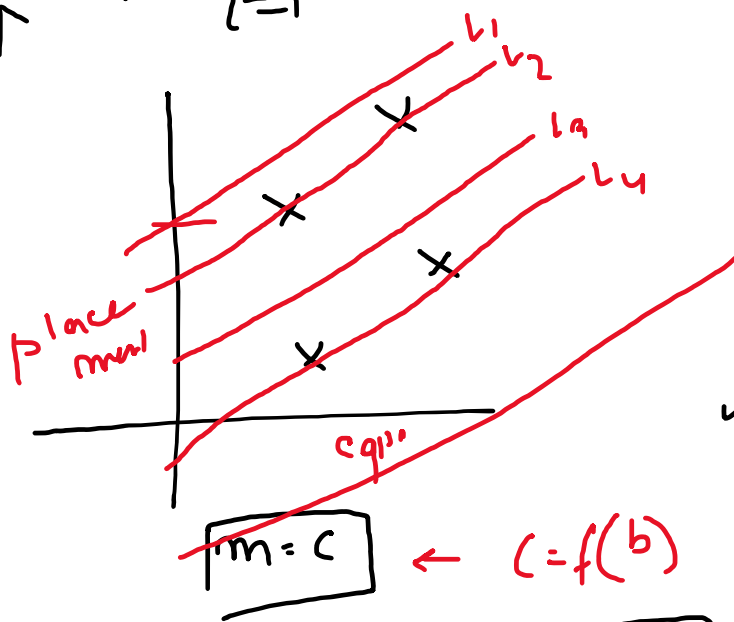
$$\begin{aligned} y &= ax + b \\ y &= f(x) \end{aligned}$$

MSR

$$C = \frac{1}{n} \sum_{i=1}^n (y_i - mx_i - b)^2$$

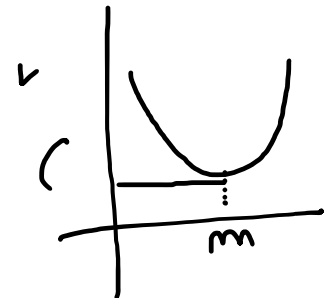
LL

$$C = \bar{C} = \frac{1}{n} \sum_{i=1}^n C_i$$



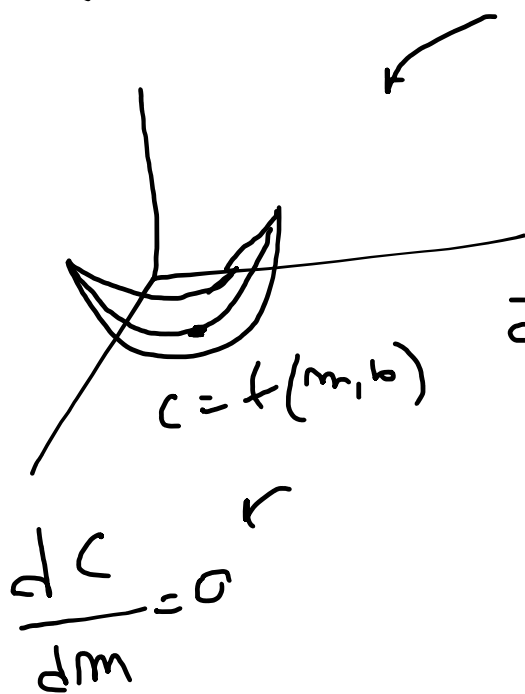
$$\frac{dC}{db} = 0$$

Assignment $C = f(m)$
 $b \leftarrow \text{const}$

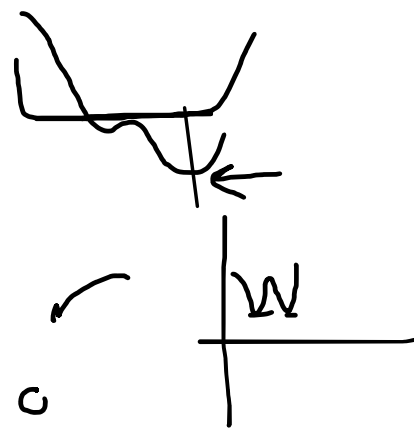


$$\frac{dC}{dm} = 0$$

$$C = f(m, b)$$



$$\frac{\partial^2 y}{\partial x^2}$$



$$\frac{\partial C}{\partial b} = 0$$

~~place~~

$$\Rightarrow \sum_{i=1}^n [y_i - \hat{y}_i]^2 = 0$$

$$\frac{d}{dm}$$

$$\Rightarrow \frac{d}{dm} \left[\frac{1}{n} \sum_{i=1}^n (y_i - m x_i - b)^2 \right] = 0$$

$$\Rightarrow \frac{d}{db} \left[\frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2 \right] = 0$$

$$\Rightarrow \frac{d}{db} \left[\frac{1}{n} \sum_{i=1}^n (y_i - m x_i - b)^2 \right] = 0$$

$$\Rightarrow \frac{d}{dm} \frac{1}{n} \sum_{i=1}^n (y_i - m x_i - \bar{y} + m \bar{x})^2 = 0$$

$$\Rightarrow \frac{1}{n} \sum_{i=1}^n 2 (y_i - m x_i - b) x_i = 0$$

$$\Rightarrow \frac{1}{n} \sum_{i=1}^n 2 (y_i - m x_i - \bar{y} + m \bar{x}) (-x_i + \bar{x})$$

$$\Rightarrow \frac{1}{n} \sum_{i=1}^n 2 (y_i - m x_i - b) = 0$$

$$\Rightarrow \frac{1}{n} \sum_{i=1}^n (y_i - m x_i - \bar{y} + m \bar{x}) (x_i - \bar{x}) = 0$$

$$\Rightarrow + \frac{1}{n} \sum_{i=1}^n (y_i - m x_i - b) = 0$$

$$\Rightarrow \sum_{i=1}^n \left[(y_i - \bar{y}) - m (x_i - \bar{x}) \right] (x_i - \bar{x}) = 0$$

$$\Rightarrow \frac{1}{n} \sum_{i=1}^n y_i - \frac{1}{n} \sum_{i=1}^n m x_i - \frac{1}{n} \sum_{i=1}^n b = 0$$

$$\Rightarrow \frac{\sum (y_i - \bar{y}) (x_i - \bar{x})}{\sum (x_i - \bar{x})^2}$$

$$\Rightarrow 1. \frac{(y_1 + y_2 + \dots + y_n)}{n}$$

$$1. m \frac{x_1 + x_2 + \dots + x_n}{n}$$

$$\frac{b + b + \dots + b}{n} = 0$$

$$\bar{y} - m \bar{x} - b = 0$$

$$b = \bar{y} - m \bar{x}$$

$\bar{y} \leftarrow \text{mean of } y$

$$m = \frac{\sum (y_i - \bar{y}) (x_i - \bar{x})}{\sum (x_i - \bar{x})^2}$$

num = 0
row 1 $\rightarrow (y_1 - \bar{y}) (x_1 - \bar{x}) = \text{num}$
row 2 $\rightarrow (y_2 - \bar{y}) (x_2 - \bar{x}) + \text{num}$
num

row 1 $(y_1 - \bar{y}) / (x_1 - \bar{x}) \rightarrow$ num
 row 2 $(y_2 - \bar{y}) / (x_2 - \bar{x}) \rightarrow$ den

✓

$\bar{y} \leftarrow$ mean of y
 $\bar{x} \leftarrow$ mean of x

$$\hat{y} = mx + b$$

$$(y_i - \bar{y}) / (x_i - \bar{x})$$

$m = \textcircled{1}$
 $b = \textcircled{1}$

$$X_{\text{test}} = \begin{bmatrix} [\bar{x}] \\ [1] \\ [] \\ [] \end{bmatrix}$$

$m \cdot X_{\text{test}} + b$

$$\begin{bmatrix} [] \\ [] \\ [] \end{bmatrix} \Rightarrow \begin{bmatrix} [] \\ [] \\ [] \end{bmatrix}$$

✓ $\overline{MAE} \Rightarrow$ mean Absolute Error $\rightarrow$ Unit Same

$\rightarrow$ $MSE \Rightarrow$ " Squared Error $\rightarrow$ sq

✓ $\overline{gpa}$ (LPA) $\leftarrow$ placement MAE $C = \frac{1}{n} \sum_{i=1}^n y_i - \hat{y}_i \leftarrow (LPA)$

$MSE \quad C = \frac{1}{n} \sum (y_i - \hat{y}_i)^2 \leftarrow (LPA)^2$

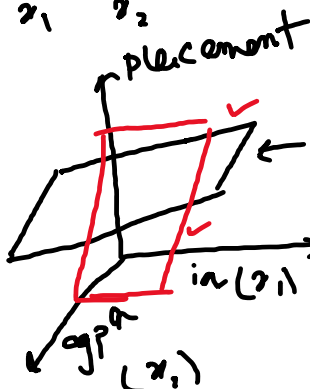
$\rightarrow$ $RMSE = C = \sqrt{\frac{1}{n} \sum (y_i - \hat{y}_i)^2} \leftarrow$ same for

← same unit

cgpa / placement ← SLR

$\frac{m}{w}$

ia / cgpa / placement
 x_1 x_2 y



①
 $y = mx + b$ (SLR)
 $y = w_1x_1 + w_2x_2 + w_0$

	x_1	x_2	x_3	y
row	ia	cgpa	gender	placement
1	80	7	F	8.9
2	50	5	M	0
3	90	9	F	12
4	70	8	M	8.3

row
 i, j col

$$y = w_0 + w_1x_1 + w_2x_2 + w_3x_3$$

	x_1	x_2	x_3	y
	ia	cgpa	gender	placement
1	x_{11}	x_{12}	x_{13}	
2	x_{21}	x_{22}	x_{23}	
3	x_{31}	x_{32}	x_{33}	
4	x_{41}	x_{42}	x_{43}	

$$\hat{y}_1 = w_0 + w_1x_{11} + w_2x_{12} + w_3x_{13}$$

$$\hat{y}_2 = w_0 + w_1x_{21} + w_2x_{22} + w_3x_{23}$$

$$\hat{y}_3 = w_0 + w_1x_{31} + w_2x_{32} + w_3x_{33}$$

⋮

$$\hat{Y} = \begin{bmatrix} \hat{y}_1 \\ \hat{y}_2 \\ \hat{y}_3 \\ \vdots \end{bmatrix} = \begin{bmatrix} w_0 + w_1x_{11} + w_2x_{12} + w_3x_{13} \\ w_0 + w_1x_{21} + w_2x_{22} + w_3x_{23} \\ w_0 + w_1x_{31} + w_2x_{32} + w_3x_{33} \\ \vdots \end{bmatrix}$$

$$= \begin{bmatrix} 1 & x_{11} & x_{12} & x_{13} \\ 1 & x_{21} & x_{22} & x_{23} \\ 1 & x_{31} & x_{32} & x_{33} \end{bmatrix} \begin{bmatrix} w_0 \\ w_1 \\ w_2 \\ w_3 \end{bmatrix} \leftarrow 3 \text{ row} \times 3 \text{ col}$$

$$= \begin{bmatrix} 1 & x_{11} & x_{12} & x_{13} \\ 1 & x_{21} & x_{22} & x_{23} \\ \vdots & \vdots & \vdots & \vdots \\ 1 & x_{m1} & x_{m2} & x_{m3} \end{bmatrix} \begin{bmatrix} w_0 \\ w_1 \\ w_2 \\ w_3 \end{bmatrix}$$

m row x 3 col

w_0 w_1 w_2 w_3 y
 x_0 x_1 x_2 x_3 y
 1 x_1 x_2 x_3 y

$$= \begin{bmatrix} 1 & x_{11} & x_{12} & x_{13} & \dots & x_{1m} \\ 1 & x_{21} & x_{22} & x_{23} & \dots & x_{2m} \\ \vdots & \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & x_{m1} & x_{m2} & x_{m3} & \dots & x_{mm} \end{bmatrix} \begin{bmatrix} w_0 \\ w_1 \\ \vdots \\ w_n \end{bmatrix}$$

m row x m col

$$\hat{Y} = X \cdot W$$

$$y = w_0 + w_1 x_1 + w_2 x_2 + \dots + w_n x_n$$

$$= \underline{w_0 x_0} + w_1 x_1 + \dots + w_n x_n \quad x_0 = 1$$

$$y = \sum_{i=0}^n w_i x_i \quad \text{where } x_0 = 1$$

$$\hat{Y} = \underline{X \cdot W}$$

$$Y = \begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_m \end{bmatrix} \quad \hat{Y} = \begin{bmatrix} \hat{y}_1 \\ \hat{y}_2 \\ \vdots \\ \hat{y}_m \end{bmatrix}$$

$$e = (Y - \hat{Y}) \leftarrow \text{matrix form}$$

msfe

$$E = \frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2$$

$$B = f(w_0, w_1, w_2, \dots, w_n)$$

to minimize

function

$$E = \sum_{i=1}^n (y_i - \hat{y}_i)^2$$

matrix

$$E \rightarrow \begin{bmatrix} (y_1 - \hat{y}_1)^2 \\ (y_2 - \hat{y}_2)^2 \\ \vdots \\ (y_n - \hat{y}_n)^2 \end{bmatrix}$$

math

$$= (y_1 - \hat{y}_1)^2 + (y_2 - \hat{y}_2)^2 + \dots + (y_n - \hat{y}_n)^2$$

$$E = e^T e \Rightarrow e = \begin{bmatrix} y_1 - \hat{y}_1 \\ y_2 - \hat{y}_2 \\ \vdots \\ y_n - \hat{y}_n \end{bmatrix}$$

$$e^T = \text{transpose of } e$$

$$= \begin{bmatrix} (y_1 - \hat{y}_1) & \dots & (y_n - \hat{y}_n) \end{bmatrix}$$

$$\begin{bmatrix} (y_1 - \hat{y}_1) & (y_2 - \hat{y}_2) & \dots & (y_n - \hat{y}_n) \end{bmatrix}_{1 \times n} \times \begin{bmatrix} y_1 - \hat{y}_1 \\ y_2 - \hat{y}_2 \\ \vdots \\ y_n - \hat{y}_n \end{bmatrix}_{n \times 1}$$

$$= \begin{bmatrix} y_1 - \hat{y}_1 \\ y_2 - \hat{y}_2 \\ \vdots \\ y_n - \hat{y}_n \end{bmatrix}_{n \times 1} \begin{bmatrix} (y_1 - \hat{y}_1) & (y_2 - \hat{y}_2) & \dots & (y_n - \hat{y}_n) \end{bmatrix}_{1 \times n}$$

$$\hat{Y} = XW$$

$$\begin{aligned}
 \rightarrow E &= e^T e \\
 &= (Y - \hat{Y})^T (Y - \hat{Y}) \quad \left| \begin{array}{l} (A-B)^T \\ = A^T - B^T \end{array} \right. \\
 &= (Y^T - \hat{Y}^T) (Y - \hat{Y}) \\
 &= (Y^T - (XW)^T) (Y - XW) \\
 &= Y \cdot Y^T - \underbrace{Y^T XW - Y (XW)^T}_{\boxed{Y^T XW = Y (XW)^T}} + XW (XW)^T
 \end{aligned}$$

$$= YY^T - 2Y^T XW + XW (XW)^T$$

$$\boxed{f' = Y\hat{Y} - 2Y^T XW + XW (XW)^T}$$

$$\boxed{\frac{dE}{dm} \quad \frac{dE}{dh}}$$

$$\begin{aligned}
 \frac{dE}{dW} &= \frac{d}{dW} \left[Y\hat{Y} - 2Y^T XW + XW (XW)^T \right] = 0 \\
 &= 0 - 2Y^T X + \frac{d}{dW} \underbrace{XW}_{\downarrow} \underbrace{X^T}_{\uparrow} W^T = 0
 \end{aligned}$$

$$y = A^T \cdot A$$
$$\frac{dy}{dx} = 2 \cdot A^T$$
$$\Rightarrow -2Y^T X + 2 \underline{x}^T \underline{X}^w = 0$$
$$\underline{X}^w = \underline{X}^T$$

$$\Rightarrow \cancel{X^T} X W^T = \cancel{Y^T} X$$

$$\Rightarrow (W^T)^T = (Y^T X (X^T X)^{-1})^T$$

$$\Rightarrow W = Y X^T (X^T X)^{-1}$$

$$W = Y X^T (X^T X)^{-1}$$

←
[$w_0 \ w_1 \dots w_n$]

$$\underline{MLR} = E = \frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2$$

$$y_i = \sum_{j=0}^3 w_j x_j \quad \leftarrow \boxed{x_0 = 1}$$

$$\text{SLN} \rightarrow \hat{y}_i = w_0 + w_1 x_i = \sum_{i=0}^1 w_i x_i$$

$$Y =$$

$$X \quad W \quad \gamma \quad \left[w_c \right]$$

$$\begin{aligned}
 \hat{Y} &= \begin{bmatrix} 1 & x_{11} & x_{12} & x_{13} & \dots \\ 1 & x_{21} & \dots & \dots & \dots \\ \vdots & \vdots & \vdots & \vdots & \vdots \\ 1 & x_{m1} & \dots & \dots & \dots \end{bmatrix} \begin{bmatrix} w_0 \\ w_1 \\ \vdots \\ w_n \end{bmatrix} \\
 &= \begin{bmatrix} w_0 + w_1 x_{11} + w_2 x_{12} + \dots \\ w_0 + w_1 x_{21} + \dots \\ \vdots \\ w_0 + w_1 x_{m1} + \dots \end{bmatrix} \begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_m \end{bmatrix} \\
 &= \begin{bmatrix} w_0 \\ \vdots \\ w_n \end{bmatrix} + \begin{bmatrix} x_{11} & x_{12} & \dots \\ \vdots & \vdots & \vdots \\ x_{m1} & x_{m2} & \dots \end{bmatrix} \begin{bmatrix} w_1 \\ w_2 \\ \vdots \\ w_n \end{bmatrix}
 \end{aligned}$$

$$= w_0 + X \cdot W$$

$$\hat{Y} = \text{intercept} + X \cdot \text{weight} \text{ (coeff)}$$

OLS ←

$$\text{Matrix Diff}^n = O(n^3)$$

$$\text{fct} \rightarrow \text{Diff}^n$$